

WAWA, ON, Canada

WMO: 717380

Lat: 47.9730N

Lon: 84.7824W

Elev: 942

StdP: 14.20

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.0	-18.0	-30.6	1.0	-23.7	-24.8	1.4	-17.5	22.8	22.1	20.6	22.4	2.2	20	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.5	77.5	65.7	74.5	64.2	72.1	62.8	68.6	73.7	66.8	71.8	65.1	69.8	8.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.8	102.3	71.2	64.9	95.5	69.3	63.2	89.9	67.9	33.5	73.7	31.9	72.1	30.6	69.9	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.4	14.3	-32.1	83.5	5.2	2.4	-35.8	85.3	-38.8	86.7	-41.7	88.1	-45.5	89.8		
(5)			-32.6	72.7	5.0	1.6	-36.2	73.8	-39.1	74.7	-42.0	75.6	-45.6	76.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.8	8.2	11.1	21.1	34.0	46.8	55.9	59.6	60.2	54.0	42.3	29.6	16.9		
	DBStd	20.76	14.10	12.60	12.45	8.48	7.87	6.61	5.51	5.23	7.48	8.03	10.44	12.83		
	HDD50	5898	1296	1089	896	482	158	20	3	1	44	269	614	1026		
	HDD65	10348	1761	1509	1361	930	566	279	180	166	337	705	1063	1491		
	CDD50	1071	0	0	0	2	59	199	302	316	163	29	1	0		
	CDD65	45	0	0	0	0	2	7	14	16	6	0	0	0		
	CDH74	300	0	0	0	0	35	76	85	77	26	1	0	0		
	CDH80	24	0	0	0	0	3	7	6	7	1	0	0	0		
	WSAvg	5.7	6.0	6.2	6.0	6.2	5.6	4.9	4.6	5.1	5.8	6.1	6.3	6.0		
	PrecAvg	40.2	3.1	2.2	2.2	2.8	3.3	3.1	3.4	3.3	4.8	4.6	3.5	3.7		
(15) Precipitation	PrecMax	55.2	6.1	6.1	4.1	5.9	8.3	6.3	6.4	7.5	10.3	8.4	6.2	8.0		
	PrecMin	19.0	1.6	0.7	0.3	0.6	1.0	0.7	0.5	0.7	1.5	1.5	1.1	1.6		
	PrecStd	8.7	1.0	1.1	0.9	1.5	1.6	1.5	1.5	1.6	2.1	2.0	1.2	1.4		
	PrecStd	8.7	1.0	1.1	0.9	1.5	1.6	1.5	1.5	1.6	2.1	2.0	1.2	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.4	43.7	56.2	65.6	78.7	80.7	80.6	80.2	77.6	69.5	55.5	42.5		
		MCWB	36.3	40.2	46.3	50.9	64.3	65.0	67.4	68.3	66.1	60.7	50.7	40.8		
	2%	DB	35.2	36.8	47.3	57.1	71.7	75.9	76.5	75.8	72.9	64.4	50.8	39.1		
		MCWB	33.8	34.5	41.2	45.0	57.9	63.3	65.8	66.3	64.6	57.5	47.3	37.2		
	5%	DB	32.9	33.2	41.8	51.7	66.6	72.6	73.3	72.9	69.7	60.1	47.2	36.7		
		MCWB	31.7	31.4	37.0	42.6	54.9	61.2	64.1	64.6	63.4	55.4	44.0	35.1		
	10%	DB	29.8	30.3	37.5	47.5	61.9	69.4	70.7	70.3	66.5	55.6	43.7	34.2		
		MCWB	28.5	28.6	33.8	40.1	52.3	59.2	62.8	63.1	61.4	51.8	41.1	32.7		
	0.4%	WB	36.7	40.9	48.6	52.1	66.3	69.9	70.6	71.2	69.0	62.7	52.2	41.3		
		MCDB	37.6	42.4	51.9	60.5	75.4	75.8	76.3	76.2	72.9	66.6	55.0	41.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.0	35.0	41.9	47.4	61.3	66.6	68.1	68.4	66.5	58.5	48.1	37.6		
		MCDB	35.0	36.5	46.0	54.3	68.7	72.4	73.3	73.2	71.0	63.1	50.5	38.7		
	5%	WB	31.9	31.8	37.5	44.1	57.2	63.8	66.1	66.4	64.5	55.6	44.2	35.1		
		MCDB	32.8	33.0	40.7	50.6	63.6	69.3	71.1	71.1	68.2	59.6	46.6	36.5		
	10%	WB	28.4	28.6	34.1	40.5	53.6	61.1	64.2	64.7	62.1	52.0	41.3	32.9		
		MCDB	29.7	30.3	36.8	46.5	59.5	66.7	69.0	68.8	65.7	55.8	43.6	34.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.5	20.7	21.1	19.3	21.6	21.3	18.3	17.5	17.0	14.5	12.9	15.6		
		MCDBR	13.1	17.9	20.8	26.9	29.7	26.5	22.4	20.9	19.8	20.5	15.8	13.7		
		MCWBR	12.8	16.0	14.5	16.7	17.0	14.9	13.1	12.6	12.5	14.5	13.2	12.8		
		MCDBR	13.2	16.9	18.4	24.5	24.5	21.7	18.8	17.5	17.1	18.2	14.8	13.1		
	5% WB	MCWBR	13.2	15.6	14.2	16.7	16.2	14.6	12.7	11.7	12.3	14.1	13.2	12.6		
		taub	0.273	0.271	0.308	0.335	0.360	0.377	0.387	0.367	0.365	0.325	0.290	0.297		
		taud	2.309	2.319	2.320	2.388	2.404	2.380	2.369	2.428	2.433	2.529	2.522	2.237		
		Ebn at Noon	255	285	289	291	286	281	276	277	267	263	252	224		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	352	652	1099	1484	1727	1852	1843	1553	1125	666	333	249		
		RadStd	51	76	106	145	141	123	129	113	78	85	42	34		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page